

Superstore Business Analysis Report

Performance Evaluation, Profitability Analysis & Operational Insights — 2019 Dataset

Executive Summary

This report analyzes the Superstore 2019 dataset to evaluate overall business performance and identify areas that may require management attention.

The business generated **\$2.30 million** in sales and **\$286.40 thousand** in profit, resulting in an overall profit margin of **12.47%**.

While the overall business is profitable, the analysis reveals several important performance differences beneath the total numbers:

- **26.31%** of unique orders were **unprofitable**.
- Orders with discounts above 20% generated a combined loss of **-\$135,376.06**.
- High revenue does not always translate into high profitability. The clearest example is the **Tables** sub-category, which generated approximately \$206,966 in sales while producing a loss of **-\$17,725**.

These findings highlight several areas that deserve further investigation, particularly discounting, product profitability, and the relationship between sales volume and actual profit contribution.

1. Business Performance Overview

The first step in evaluating the business is to understand its overall financial performance.

Measure	Result
Total Sales	\$2,297,200.86
Total Profit	\$286,397.02
Profit Margin	12.47%
Average Order Value	\$458.61
Unprofitable Orders	26.31%

The business generated positive aggregate profit. However, the 26.31% unprofitable-order ratio shows that the positive overall result does not mean that every part of the business is profitable. This distinction is important because overall performance can hide specific products, transactions, or customer groups that are reducing profitability.

2. Sales Versus Profitability

A comparison between sales and profit reveals an important pattern: the highest sales areas are not necessarily the strongest profitability areas.

Category	Sales	Profit	Profit Margin
Technology	\$836,154.03	\$145,454.95	17.39%
Furniture	\$741,999.80	\$18,451.27	2.49%
Office Supplies	\$719,047.03	\$122,490.80	17.03%

Technology generated both the highest aggregate sales and the highest aggregate profit. Furniture, however, generated almost \$742K in sales while producing only \$18.5K in profit (a slim 2.49% margin). This difference suggests that Furniture deserves a closer review of pricing, discounts, product mix, and costs. The data alone does not identify which of these factors is responsible, so further investigation would be required before making an operational change.

3. Product-Level Profitability

Sub-category analysis provides a more detailed view of where profitability is being generated or lost.

Sub-Category	Sales	Profit	Status
Tables	\$206,965.53	-\$17,725.48	High Loss
Bookcases	\$114,879.99	-\$3,472.56	Unprofitable
Chairs	\$328,449.10	\$26,590.17	Profitable
Phones	\$330,007.05	\$44,515.73	Profitable
Copiers	\$149,528.03	\$55,617.82	High Margin

Detailed Sub-category Insights:

- **Tables:** Represent one of the clearest profitability concerns in the dataset. The sub-category generated more than \$206K in sales but resulted in a loss of approximately \$17.7K. A business review could investigate discount levels, product-level margins, product costs, customer segments, geographic performance, shipping costs, and return behavior.
- **Bookcases:** Generated a negative aggregate profit despite producing more than \$114K in sales. This reinforces the need to evaluate product profitability rather than relying only on revenue.
- **Copiers:** Generated approximately \$149.5K in sales and \$55.6K in profit, representing a substantially different profitability pattern from Tables and illustrating why product-level analysis can reveal information hidden by total sales.

4. Discount Analysis

Key Discounting Finding:

Orders with discounts above 20% generated **-\$135,376.06** in combined profit.

This result indicates a strong negative association between the high-discount group and aggregate profitability in this dataset. However, the analysis does not establish causation. For example, high discounts may be concentrated in specific products, categories, customers, or circumstances that independently have lower profitability.

Therefore, before changing the discount policy, management would benefit from examining profit margins by discount level, category, sub-category, product, customer segment, region, and total cost structure.

5. Unprofitable Orders

The analysis found that **26.31%** of unique orders were unprofitable. This is significant from a management perspective because the business can maintain positive total profit while a considerable share of individual orders generate losses.

A useful follow-up analysis would divide these unprofitable orders by Category, Sub-category, Product, Discount level, Customer segment, Region, and Shipping mode to identify where losses are concentrated.

6. Customer Segment Performance

Customer Segment	Sales	Profit
Consumer	~\$1,160,000.00	\$134,119.21
Corporate	\$706,146.40	\$91,979.13
Home Office	\$429,653.10	\$60,298.68

Consumer customers generated the highest aggregate sales and profit. However, this result should be interpreted as total contribution rather than profitability per customer. A more detailed customer analysis would examine average customer revenue, average customer profit, order frequency, repeat purchasing, customer lifetime value, and loss-making customer accounts.

7. Performance Over Time

Year	Sales	Profit	Growth / Notes
2016	\$484,247.50	\$49,543.97	Baseline Year
2017	\$470,532.51	\$61,618.60	+24.4% Profit Growth
2018	\$609,205.60	\$81,795.17	+32.7% Profit Growth
2019	\$733,215.26	\$93,439.27	+14.2% Profit Growth

The business experienced substantial growth in both sales and profit between 2017 and 2019. Sales increased from approximately \$470.5K in 2017 to \$733.2K in 2019. Profit increased from approximately \$61.6K to \$93.4K during the same period.

8. Geographic Performance

City	Sales	Profit
New York City	\$256,368.16	\$62,036.98
Los Angeles	\$175,851.34	\$30,440.76

City	Sales	Profit
Seattle	\$119,540.74	\$29,156.10
San Francisco	\$112,669.09	\$17,507.39
Philadelphia	\$109,077.01	Not provided

New York City generated the highest aggregate sales and profit among the cities included in the provided results. For management purposes, geographic differences could be investigated alongside product mix, discounts, customer segments, and regional costs.

9. Shipping Performance

Shipping Mode	Sales	Profit
Standard Class	\$1,358,216.00	\$164,088.79
Second Class	\$459,193.60	\$57,446.64
First Class	\$351,428.40	\$48,969.84
Same Day	\$128,363.10	\$15,891.76

Standard Class generated the largest aggregate sales and profit. However, this should not be interpreted as evidence that Standard Class is inherently more profitable, as the result is affected by order volume. A detailed evaluation would compare profit per order, average shipping transit days, and shipping modes across product categories.

10. Business Areas Requiring Attention

1. Discount Management:

Business Question: Are discounts reducing profitability across the business, or is the effect concentrated in specific products or categories?

2. Loss-Making Products:

Business Question: What factors are causing Tables and Bookcases to sell at a loss?

3. Furniture Profitability:

Business Question: Is the profitability gap driven by discounting, product costs, product mix, or specific sub-categories?

4. Unprofitable Orders:

Business Question: Where are these 26.31% loss-making orders concentrated?

5. Customer Profitability:

Business Question: Which customer groups generate the strongest profit relative to their order volume and sales?

6. Geographic Differences:

Business Question: Are geographic differences driven by customer mix, product mix, pricing, or discounting?

11. Recommended Analytical Follow-Up

- **Product-Level Analysis:** Identify the individual products responsible for losses within Tables, Bookcases, and other low-profit areas.
- **Discount-Margin Analysis:** Compare profit margins across discount levels while separating categories and products.
- **Loss-Making Order Analysis:** Identify the common characteristics shared by unprofitable orders.
- **Customer Profitability:** Calculate customer-level profit and compare it with sales and order frequency.
- **Return Impact:** Measure whether returned orders or specific product groups are associated with lower profitability.
- **Shipping Economics:** Compare shipping modes using profit per order and other operational measures rather than aggregate totals alone.

12. Important Interpretation Notes

- Association does not prove causation.
- Aggregate sales and profit can hide differences between individual orders.
- External operating costs may not be included in the dataset.
- High-value transactions can strongly affect aggregate results.
- The dataset does not provide all business context needed for operational decisions.
- Additional analysis is required before changing pricing, discount, product, or shipping policies.

13. Conclusion

The analysis shows a business with positive overall profitability and clear growth in sales and profit over the analyzed period. At the same time, the results reveal several areas that deserve closer management attention, notably \$2.30M in total sales, \$286.40K in profit (12.47% margin), but balanced against 26.31% unprofitable orders and over -\$135K in losses from orders discounted above 20%.

The main business lesson from the analysis is that revenue alone does not provide a complete picture of business performance. Combining sales with profit, margins, discounts, products, customers, locations, and order-level behavior provides a more useful basis for identifying where the business should investigate further.